



jQuery Cheat Sheet

23 Essential Questions with Examples and Best Practices

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What is jQuery and why is it used?

jQuery is a fast, small, and feature-rich JavaScript library that simplifies:

- HTML DOM traversal
- Event handling
- AJAX interactions
- Animations

```
$("#myDiv").hide(); // Hides an element

<button id="btnHide">Hide</button>
<div id="box">Hello World</div>

<script>
$("#btnHide").click(function() {
    $("#box").hide(); // hides the div
});
</script>
```

What is the use of `$(document).ready()`?

Ensures the DOM is fully loaded before executing jQuery code:

```
$(document).ready(function() {  
    // DOM is ready for manipulation  
    $("#content").show();  
});  
  
// Shorthand version:  
$(function() {  
    $("button").css("background", "#3498db");  
});
```

How do you select elements?

```
// By ID  
$("#header").text("Welcome!");  
  
// By class  
$(".active").removeClass("active");  
  
// By tag  
$("p").css("color", "blue");  
  
// By attribute  
$("a[target='_blank']").addClass("external");
```

What is method chaining?

Execute multiple methods on the same selection:

```
// With chaining
$("#myDiv")
  .css("color", "blue")
  .slideUp(1000)
  .slideDown(1000);

// Without chaining
$("#myDiv").css("color", "blue");
$("#myDiv").slideUp(1000);
$("#myDiv").slideDown(1000);
```

How to show/hide elements?

```
// Basic show/hide
$("#box").hide();
$("#box").show();

// With effects
$("#box").fadeOut(1000); // 1 second fade
$("#box").fadeIn(1000);
$("#box").slideUp(500);
$("#box").slideDown(500);
```

How to attach multiple event handlers?

```
// Chain .on() for multiple events
$("#myBtn")
  .on("click", function() {
    alert("Clicked!");
  })
  .on("mouseenter", function() {
    $(this).css("background", "lightblue");
  });

// Pass multiple events as an object
$("#myBtn").on({
  click: function() {
    alert("Button clicked");
  },
  mouseenter: function() {
    $(this).css("background-color", "yellow");
  },
  mouseleave: function() {
    $(this).css("background-color", "");
  }
});
```

How to prevent default event behavior?

```
<form id="myForm">
  <input type="text" required>
  <button type="submit">Submit</button>
</form>

$("#myForm").on("submit", function(e) {
  e.preventDefault(); // Stop form submission

  if ($("#input").val() === "") {
    alert("Field cannot be empty!");
  } else {
    // Process form
  }
});
```

How to stop event bubbling?

```
<div id="box">
  <button id="btn">Click Me</button>
</div>

$("#box").click(function() {
  alert("Box clicked!");
});

$("#btn").click(function(e) {
  e.stopPropagation(); // Prevent event from bubbling up
  alert("Button clicked!");
});
```

What happens?

- Only the button's alert will show
- The box's alert will not be triggered

What is event delegation?

Attach an event to a parent element, but handle events for child elements (even if they are added later).

```
<ul id="myList">
  <li>Item 1</li>
  <li>Item 2</li>
</ul>

// Using event delegation:
$("#myList").on("click", "li", function() {
  alert("You clicked: " + $(this).text());
});

// Add new items dynamically
$("#myList").append("<li>Item 3</li>");
```

How to remove event handlers?

```
// Remove all handlers
$("#btn").off();

// Remove specific handler
$("#btn").off("click");

// Remove namespaced handler
$("#btn").on("click.myPlugin", handler);
$("#btn").off("click.myPlugin");
```

What does \$.each() do?

Loops through arrays, objects, and jQuery collections.

```
// Loop through an array
const fruits = ["Apple", "Banana", "Mango"];

$.each(fruits, function(index, value) {
    console.log(index + ": " + value);
});

// Loop through an object
const person = {
    name: "John",
    age: 30,
    occupation: "Developer"
};

$.each(person, function(key, value) {
    console.log(key + ": " + value);
});

// Loop through jQuery elements
$("li").each(function(index) {
    console.log(index + ": " + $(this).text());
});
```


Difference between .prop() and .attr()

Method	Description	Example
.attr()	Gets or sets HTML attributes	<code>\$("img").attr("src", "new.jpg")</code>
.prop()	Gets or sets DOM properties	<code>\$("checkbox").prop("checked", true)</code>

When to use which:

- Use `.attr()` for HTML attributes (src, href, class, etc.)
- Use `.prop()` for DOM properties (checked, disabled, selected, etc.)

What is \$.extend() used for?

Merges objects into the first object.

```
// Merge default options with user settings
const defaults = {
  color: "blue",
  size: "medium"
};

const userOptions = {
  color: "red",
  speed: "fast"
};

const settings = $.extend({}, defaults, userOptions);

// Result:
// {
//   color: "red",
//   size: "medium",
//   speed: "fast"
// }
```

Difference between \$(this) and this

Type	Description	Example
this	JavaScript keyword (raw DOM element)	this.style.color = "red"
\$(this)	jQuery object wrapping this	\$(this).css("background", "yellow")

```
<button class="btn">Click Me</button>

$(".btn").click(function() {
  // Raw DOM element
  console.log(this);
  this.style.color = "red";

  // jQuery object
  console.log($(this));
  $(this).css("background", "yellow");
});
```

What are jQuery effects?

Built-in methods that add interactivity or animations.

```
// Basic effects
$("#box").hide();
$("#box").show();
$("#box").toggle();

// Fading effects
$("#box").fadeIn(1000);
$("#box").fadeOut(1000);
$("#box").fadeTo(1000, 0.5); // Fade to 50% opacity

// Sliding effects
$("#box").slideDown(500);
$("#box").slideUp(500);
$("#box").slideToggle(500);

// Custom animations
$("#box").animate({
  left: "250px",
  opacity: "0.5",
  height: "150px"
}, 1000);
```

Difference between .eq() and :eq()

Method	Description	Example
<code>.eq()</code>	Method that filters a jQuery collection	<code>\$('#div').eq(2).addClass('highlight')</code>
<code>:eq()</code>	Selector filter (pseudo-class)	<code>\$('#div:eq(2)').addClass('highlight')</code>

Note: Both are 0-indexed but `:eq()` is a selector filter, while `.eq()` is a method applied to an existing jQuery object.

How to handle form submission?

```
$("#myForm").on("submit", function(e) {  
    e.preventDefault(); // Prevent page reload  
  
    // Form validation  
    const email = $("#email").val();  
    if (!email.includes("@")) {  
        alert("Invalid email!");  
        return;  
    }  
  
    // Show loading indicator  
    $("#spinner").show();  
  
    // Send AJAX request  
    $.ajax({  
        url: "submit.php",  
        method: "POST",  
        data: $(this).serialize(),  
        success: function(response) {  
            $("#result").html("Success!");  
        },  
        complete: function() {  
            $("#spinner").hide();  
        }  
    });  
});
```

How to serialize form data?

```
<form id="userForm">
  <input type="text" name="name" value="John">
  <input type="email" name="email" value="john@example.com">
  <input type="checkbox" name="subscribe" checked>
</form>
```

```
// Serialize form data
```

```
const formData = $("#userForm").serialize();
```

```
// Result: "name=John&email=john%40example.com&subscribe=on"
```

```
// Send via AJAX
```

```
$.post("process.php", formData, function(response) {
  console.log(response);
});
```

```
// Serialize as array
```

```
const formArray = $("#userForm").serializeArray();
```

```
/* Result:
```

```
[
  {name: "name", value: "John"},
  {name: "email", value: "john@example.com"},
  {name: "subscribe", value: "on"}
]
*/
```

How to load content with .load()?

```
// Load entire page
$("#result").load("content.html");

// Load specific element
$("#result").load("content.html #section2");

// With callback
$("#result").load("content.html", function() {
    alert("Content loaded!");
});

// With data parameters
$("#result").load("data.php", {
    id: 123,
    category: "books"
});
```


How to create a jQuery plugin?

```
// Plugin structure
(function($) {
    $.fn.highlight = function(options) {
        // Default settings
        const settings = $.extend({
            color: "yellow",
            backgroundColor: "black"
        }, options);

        // Apply to each element
        return this.each(function() {
            $(this).css({
                color: settings.color,
                backgroundColor: settings.backgroundColor
            });
        });
    };
})(jQuery);

// Using the plugin
$("p").highlight({
    color: "white",
    backgroundColor: "#3498db"
});
```

Difference between .find() and .children()

Method	Depth	Performance	Selector
.children()	Immediate children only	Faster	Optional
.find()	All descendants	Slower	Required

```
<div id="parent">
  <div>Child</div>
  <section>
    <div>Grandchild</div>
  </section>
</div>

// .children() selects only direct child div
$("#parent").children("div").css("color", "red");

// .find() selects all descendant divs
$("#parent").find("div").css("background", "#eee");
```

How to optimize jQuery performance?

- **Cache selectors:** Store frequently used jQuery objects
- **Use event delegation:** For dynamic elements
- **Chain methods:** Minimize DOM traversal
- **Use IDs:** When possible for faster selection
- **Limit DOM manipulation:** Batch changes

```
// Bad: Queries DOM repeatedly
for(let i=0; i<100; i++) {
    $(".items").doSomething();
}

// Good: Cache reference
const $items = $(".items");
for(let i=0; i<100; i++) {
    $items.doSomething();
}

// Better: Use document fragment
const fragment = document.createDocumentFragment();
for(let i=0; i<100; i++) {
    const item = document.createElement("div");
    item.textContent = "Item " + i;
    fragment.appendChild(item);
}
$("#container").append(fragment);
```

How is jQuery used in real-world?

- **Legacy systems:** Enterprise applications
- **WordPress:** 90% of WP sites use jQuery
- **Rapid prototyping:** MVPs and landing pages
- **E-commerce:** Dynamic product features
- **Cross-platform:** Kiosks, medical devices

```
// E-commerce quantity adjustment
$('.quantity-input').on('change', function() {
  const price = $(this).data('unit-price');
  const qty = $(this).val();
  const total = (price * qty).toFixed(2);
  $(this).closest('.product-row').find('.total-price').text('$' + total);
});

// Touchscreen kiosk menu
$('.menu-item').on('touchstart click', function(e) {
  e.preventDefault();
  const sectionId = $(this).data('section');
  $('.content-section').hide();
  $('#'+ sectionId).fadeIn();
});
```